Speech Recognition Project Plan

TensorFlow Speech Recognition Challenge / Speech Commands v0.01

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1 Dataset and task definition

The training archive contains 1-second, 16 kHz spoken-word clips from 30 word folders – `yes, no, up, down, left, right, on, off, stop, go, bed, bird, cat, dog, eight, five, four, happy, house, marvin, nine, one, seven, sheila, six, three, tree, two, wow, zero`. The evaluation target is only the 10 commands, plus the special classes `unknown` and `silence`. The `_background_noise_` folder contains long ambient recordings that should be randomly cut into 1-second segments to form silence training examples. The project will therefore study a 12-class classification task:

- **Target commands**: `yes, no, up, down, left, right, on, off, stop, go`
- **Special classes**: `unknown`, `silence`

For research experiments, the project should use the official `validation_list.txt` and `testing_list.txt` split files. This follows the reproducible protocol proposed for Speech Commands and avoids speaker leakage between train/validation/test.

2 Network architectures

The project compare three model families:

1. **KWT-1** - Transformer based model.
2. **BC-ResNet-1** - compact CNN and the main convolutional model.
3. **MatchboxNet-3x1x64** - efficient 1D time-channel separable CNN.

These models are modern enough to give competitive results on Speech Commands while remaining small enough for repeated experimentation [1, 2, 4]. A summary is given in Table 1.

Model	Default paper frontend	Parameters	Default training setup used in the literature
BC-ResNet-1 [1]	40-dim log-Mel, 30 ms window, 10 ms hop	9.2k	200 epochs; SGD with momentum 0.9; weight decay 10^{-3}; batch size 100; LR warmup to 0.1 for 5 epochs then cosine decay; dropout 0.1; time shift $[-100, 100]$ ms; background-noise mixing; SpecAugment.
KWT-1 [2, 3]	40-dim log-Mel, 30 ms window, 10 ms hop	607k	23,000 steps; AdamW; LR 10^{-3}; batch size 512; cosine schedule; 10 warmup epochs; weight decay 0.1; label smoothing 0.1; time shift, resampling, background perturbation and SpecAugment.
MatchboxNet-3x1x64 [4]	64 MFCC, 25 ms windows, 10 ms overlap, padded to 128 frames	77k	200 epochs; NovoGrad ($\beta_1 = 0.95, \beta_2 = 0.5$); max LR 0.05, min LR 0.001; warmup-hold-decay schedule; weight decay 10^{-3}; mixed precision; batch size 128 per GPU; time shift, white noise, SpecAugment and SpecCutout.

Table 1: Models and their default paper settings

3 Audio preprocessing

Spectogram frontend To isolate architectural differences, the **comparison should use one shared frontend**: 40-bin log-Mel spectrograms extracted from 1-second clips using a 30 ms window and 10 ms hop, yielding approximately 98 time frames and 40 frequency bins per example. Each clip should be padded or cropped to 1 second before feature extraction. This applies to BC-ResNet-1 and KWT-1.

MFCC frontend MatchboxNet was originally reported with a different frontend: 64 MFCC features computed from 25 ms windows with 10 ms overlap, padded to 128 feature vectors per sample [4].

3.1 Augmentation

Shared augmentation strategies are as follows:

- random time shift,
- random mixing with background-noise clips from `_background_noise_`,
- SpecAugment (time masking, frequency masking)

BC-ResNet and KWT both use time shifting, background perturbation and SpecAugment-like masking; MatchboxNet also uses time shift plus strong spectrogram masking and cutout [1, 2, 4].

4 Handling the `unknown` classes

The standard literature setup is to evaluate on 12 classes, where the remaining 20 training words are treated as `unknown` and silence is synthesized from background noise. The project compare four strategies.

Strategy A - post-hoc remapping

Train on all 31 (30 commands + `silence`) original labels, then remap any non-target prediction to `unknown` at inference time. This strategy is simple, but it is not well aligned with the final objective because the model spends capacity separating the 20 non-target words even though the final task only asks whether they belong to a single `unknown` bucket.

Strategy B - collapsed training labels

Collapse all 20 non-target words into a single `unknown` label before training and train a 12-class classifier directly. This is the most standard and task-aligned protocol, and it is the setup followed BC-ResNet, KWT and MatchboxNet for this particular task [1, 2, 4].

To prevent the `unknown` category from dominating training, non-target word classes will be rebalanced so that the total number of `unknown` examples is approximately equal to the average size of the target keyword classes. Examples will be sampled in a class-aware manner, drawing a similar number of samples from each non-target word class to preserve diversity within the `unknown` category and avoid over-representation of a few frequent classes.

Strategy C - two-stage classification

A hierarchical two-head architecture is used. A shared backbone feeds two classification heads. Head 1 performs 3-way classification into `silence`, `unknown`, and `keyword`. Head 2 performs 10-way classification over the target keywords. During training, the loss of Head 1 is computed for all samples using standard cross-entropy with the corresponding 3-class target. The loss of Head 2 is computed only for samples whose ground-truth label belongs to the 10 target keywords; for `silence` and `unknown` samples, the Head 2 loss is masked out and does not contribute to optimization. The total loss is therefore

$$\mathcal{L} = \mathcal{L}_{\text{head1}} + \lambda\, \mathbf{1}[y \in \mathcal{K}]\, \mathcal{L}_{\text{head2}},$$

where $\mathcal{K}$ denotes the set of 10 target keywords, and λ is a scalar weight controlling the contribution of the second head ($\lambda = 1$ at first, and tuned later if needed). During inference, the prediction of Head 1 is used first; Head 2 is consulted only when Head 1 predicts `keyword`.

For the hierarchical 3-way head (`silence`/`unknown`/`keyword`), class imbalance will be handled by balanced sampling minibatches so that silence, unknown, and keyword are sampled in roughly equal proportions.

Strategy D - confidence-based rejection with uniform targets

This strategy is inspired by *Deep Anomaly Detection with Outlier Exposure* [5].

Train only an 11-class classifier for the known classes (`10 commands + silence`). For non-target training words, encourage a uniform output distribution over the 11 known classes (i.e., the target label for `unknown` class becomes $[1/11, \ldots, 1/11]$). Then, classify a sample as `unknown` at inference time if

$$\max_k p(y = k \mid x) < \tau.$$

The threshold τ is selected based on the highest f1-macro score on the validation set. For fairness, the non-target word pool will use the same class-aware balancing policy as in Strategy B, so that outlier exposure remains diverse without overwhelming the target keyword classes.

Limitations

A limitation of Strategies B–D is that their sampling policies differ from the plain full-dataset pass used in Strategy A, which means the effective number of training examples processed per epoch is not equal across strategies; consequently, comparisons should be interpreted with caution.

5 Experimental protocol

The study is organized in three stages. In the first stage, each of the three model families will be trained from scratch under all four unknown-handling strategies using literature-based default settings, the official `train/validation/test` split files, and each model's standard frontend. BC-ResNet-1 and KWT-1 will use 40-bin log-Mel features, while MatchboxNet-3x1x64 will use its native 64-dimensional MFCC frontend. For Strategy D a threshold sweep will be reported to show how the validation metrics vary with τ. In this stage, all models will be trained with a minimal shared augmentation baseline consisting of random time shift and background-noise mixing, while SpecAugment is disabled.

In the second stage, augmentation will be studied only for the best unknown-handling strategy of each backbone using a $2 \times 2 \times 2$ grid (Table 2).

Table 2: Augmentation grid factors and their off/on settings.

Factor	**Off**	**On**
Time shift	—	Uniform random shift in $[-100, 100]$ ms
Background noise	—	Mixed with p=0.8; random 1 s crop from `_background_noise_`, gain $\alpha \sim U(0, 0.3)$
SpecAugment	—	One time mask (max 10 frames) + one frequency mask (max 5 bins)

As an additional follow-up experiment, the best-performing KWT Stage 2 setting will be reused and varying only the KWT hyperparameters `attention_type` $\in \{$`time`, `freq`, `both`$\}$, `depth` $= 6$, and `num_heads` $\in \{1, 2\}$. The goal of this extra study is to test alternative KWT hyperparameter settings while reducing the total number of parameters by roughly $2\times$ (since default `depth` $= 12$ [2]) for the `time` and `freq` variants.

Model selection will be based primarily on validation 12-class macro-F1. Final test-set results will report: 12-class accuracy; 12-class macro-F1; macro-F1 over the ten target commands only; per-class precision, recall, and F1; and raw and row-normalized confusion matrices for all 12 classes.

References

[1] B. Kim, M. Lee, J. Lee, S. Yang, and K. Hwang, "Broadcasted Residual Learning for Efficient Keyword Spotting," Interspeech, 2021. https://arxiv.org/abs/2106.04140

[2] A. Berg, M. O'Connor, and M. Tairum Cruz, "Keyword Transformer: A Self-Attention Model for Keyword Spotting," Interspeech, 2021. https://arxiv.org/abs/2104.00769

[3] ARM Software, "keyword-transformer" (official implementation and training scripts). https://github.com/ARM-software/keyword-transformer

[4] S. Majumdar and B. Ginsburg, "MatchboxNet: 1D Time-Channel Separable Convolutional Neural Network Architecture for Speech Commands Recognition," Interspeech, 2020. https://arxiv.org/abs/2004.08531

[5] D. Hendrycks, M. Mazeika, and T. Dietterich, "Deep Anomaly Detection with Outlier Exposure," arXiv:1812.04606, 2018. https://arxiv.org/abs/1812.04606